

28/8/26

CS 340

Lecture-13

DFA Minimization

Suppose, we are given a DFA $M = (Q, \Sigma, q_0, \delta_M, F)$. We wish to construct from M a DFA M' s.t. $L(M') = L(M)$ and M' has the minimum no. of states among all DFAs that recognize $L(M)$.

From earlier results, we know that M' is M_L , where $L = L(M)$, up to an isomorphism. The issue is to construct M' (or M_L) efficiently from M .

First step towards minimization is to remove all states in Q that are not reachable from q_0 in M . This does not change the language accepted by M .

Exercise 1: show that $L(M) = L(M')$, where M' is the DFA obtained from M by removing all unreachable states (from the start state) of M .

In light of this, we assume w.l.o.g. that M does not have any unreachable states.

Define ~~E_q~~ for each $q \in Q$, $E_q \subseteq \Sigma^*$ as

$$E_q = \{x \in \Sigma^* \mid \delta(q_0, x) = q\}$$

In other words, E_q is the set of those words on which M reaches q when started in q_0 .

clearly $\{E_q \mid q \in Q\}$ is a partition of Σ^* .

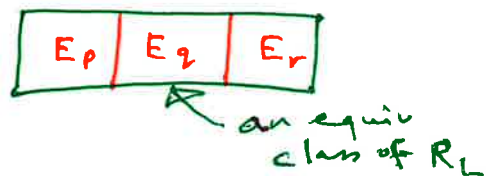
Recall relation R_M defined in the last lecture.

For $x, y \in \Sigma^*$, $x R_M y$ iff $\delta_M(q_0, x) = \delta_M(q_0, y)$

Exercise 2:

Show that partition on Σ^* induced by R_M is the same as $\{E_q \mid q \in Q\}$ defined above.

Let $L = L(M)$. From previous theory, we know that equivalence classes of R_M refine equivalence classes of R_L . For example in the figure shown alongside an equivalence class of R_L has three equivalence classes E_p, E_q, E_r of R_M .



In minimum DFA, there is only one state for this equiv. class of R_L . This suggests that we should merge states p, q, r of M . We need to identify relevant properties of these states, which allow their merger.

For $x, y \in \Sigma^*$ s.t. $x \in E_p$ and $y \in E_q$ we have

$$\forall z \in \Sigma^* [xz \in L \Leftrightarrow yz \in L] \quad (\because x R_L y)$$

$\Downarrow$

$$\forall z \in \Sigma^* [\delta(q_0, xz) \in F \Leftrightarrow \delta(q_0, yz) \in F]$$

$\Downarrow$

$$\forall z \in \Sigma^* [\delta(\delta(q_0, x), z) \in F \Leftrightarrow \delta(\delta(q_0, y), z) \in F]$$

$\Downarrow$

$$\forall z \in \Sigma^* [\delta(p, z) \in F \Leftrightarrow \delta(q, z) \in F]$$

Defn

Define a binary relation ' $\equiv$ ' on Q , states of M , as follows.

$$p \equiv q \text{ iff } \forall z \in \Sigma^* [\delta(p, z) \in F \Leftrightarrow \delta(q, z) \in F]$$

Exercise 3

observe using the derivation just before the previous definition that

$$p \equiv q \text{ iff } E_p, E_q \text{ are in the same equivalence class of } R_L$$

Exercise 4

observe that ' $\equiv$ ' defines an equivalence relation on Q . (You may use the characterization of ' $\equiv$ ' in ~~the~~ Exercise 3.)

Let $[p]$ be ~~the~~ equiv class of ' $\equiv$ ' ~~the~~ which contains state $p \in Q$. We define M' as

$$M' = (\{[p] \mid p \in Q\}, \Sigma, [q_0], \delta_{M'}, F')$$

where $\delta_{M'}([p], a) = [\delta_M(p, a)]$

and $F' = \{[p] \mid p \in F\}$

Exercise 5:

verify that $\delta_{M'}$ and F' are well defined

(Hint: use defn of ' $\equiv$ ')

Theorem:

$$L(M') = L$$

Proof:

We first show that for all $w \in \Sigma^*$

$$\delta_{M'}([q_0], w) = [\delta_M(q_0, w)]$$

This claim is ~~shown~~ shown by induction on $|w|$.

Base case: $w = \epsilon$

$$\delta_{M'}([q_0], \epsilon) = [q_0] = [\delta_M(q_0, \epsilon)]$$

Induction step:

$$w = aw_1$$

$$\begin{aligned} \delta_{M'}([q_0], aw_1) &= \delta_{M'}(\delta_{M'}([q_0], a), w_1) && \text{By def } \delta_{M'} \\ &= \delta_{M'}([\delta_M(q_0, a)], w_1) && \text{By def of } \delta_{M'} \\ &= [\delta_M(\delta_M(q_0, a), w_1)] && \text{By I.H.} \\ &= [\delta_M(q_0, aw_1)] \\ &= [\delta_M(q_0, w)] \end{aligned}$$

This proves the claim.

Now, for any $x \in \Sigma^*$,

$$x \in L(M') \Leftrightarrow \delta_{M'}([q_0], x) \in F'$$

$$\Downarrow$$

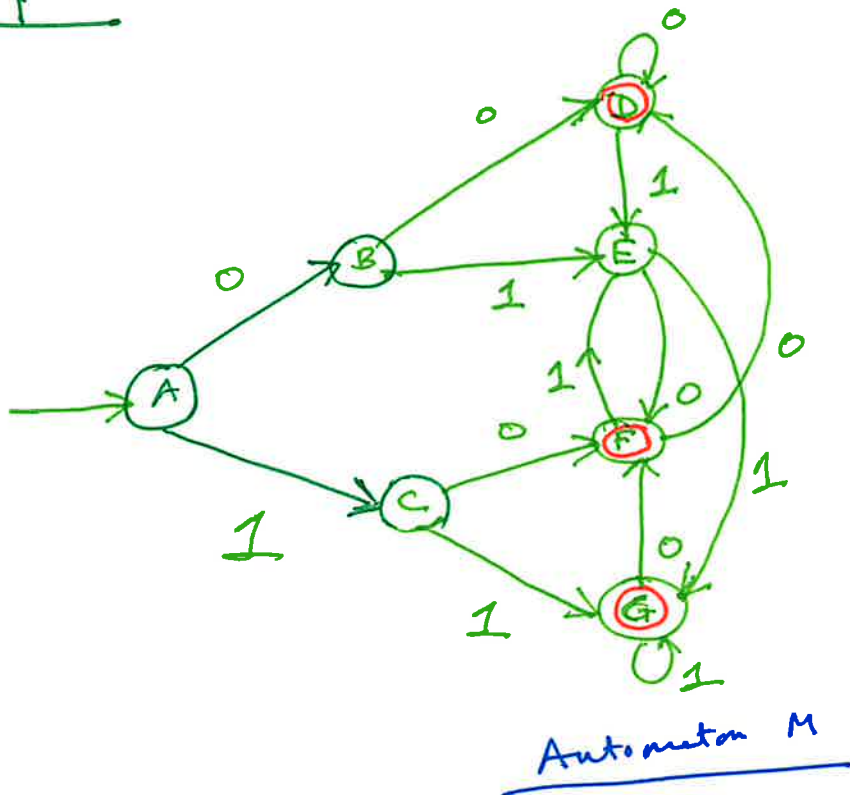
$$[\delta_M(q_0, x)] \in F'$$

$$\Downarrow$$

$$\delta_M(q_0, x) \in F \Leftrightarrow x \in L(M) \quad \square$$

(By claim above)

Example



B	X					
C	X	X				
D	X	X	X			
E	X	X		X		
F	X	X	X		X	
G	X	X	X	X	X	X
	A	B	C	D	E	F

$$(A, E) \xrightarrow{0} (B, F)$$

$$(A, C) \xrightarrow{0} (B, F)$$

$$(A, B) \xrightarrow{0} (B, D)$$

$$(G, F) \xrightarrow{0} (F, E)$$

$$(B, E) \xrightarrow{0} (D, F) \quad (B, E) \xrightarrow{1} (E, G)$$

$$(B, C) \xrightarrow{0} (D, F) \quad (B, C) \xrightarrow{1} (E, G)$$

$$(C, E) \xrightarrow{0} (F, F) \quad (C, E) \xrightarrow{1} (G, G)$$

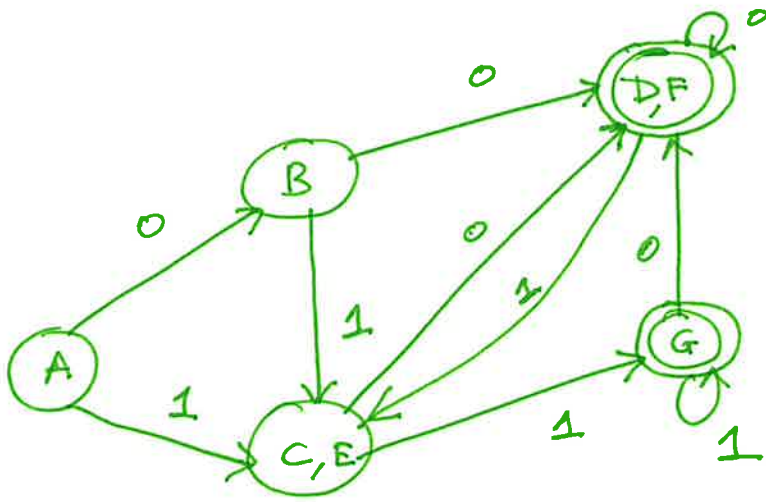
$$(D, G) \xrightarrow{0} (D, F) \quad (D, G) \xrightarrow{1} (E, G)$$

$$(D, F) \xrightarrow{0} (D, D) \quad (D, F) \xrightarrow{1} (E, E)$$

Further iterations of the table filling process, do not lead to any new state pairs ^{getting} marked. Therefore, we have final ' $\equiv$ ' relation as shown in the table.

More specifically, $C \equiv E$ and $D \equiv F$ are the only non-trivial equivalences on states of M .

We get the following minimum state automaton M' .



Automaton M'

Minimum state
DFA for $L(M)$.